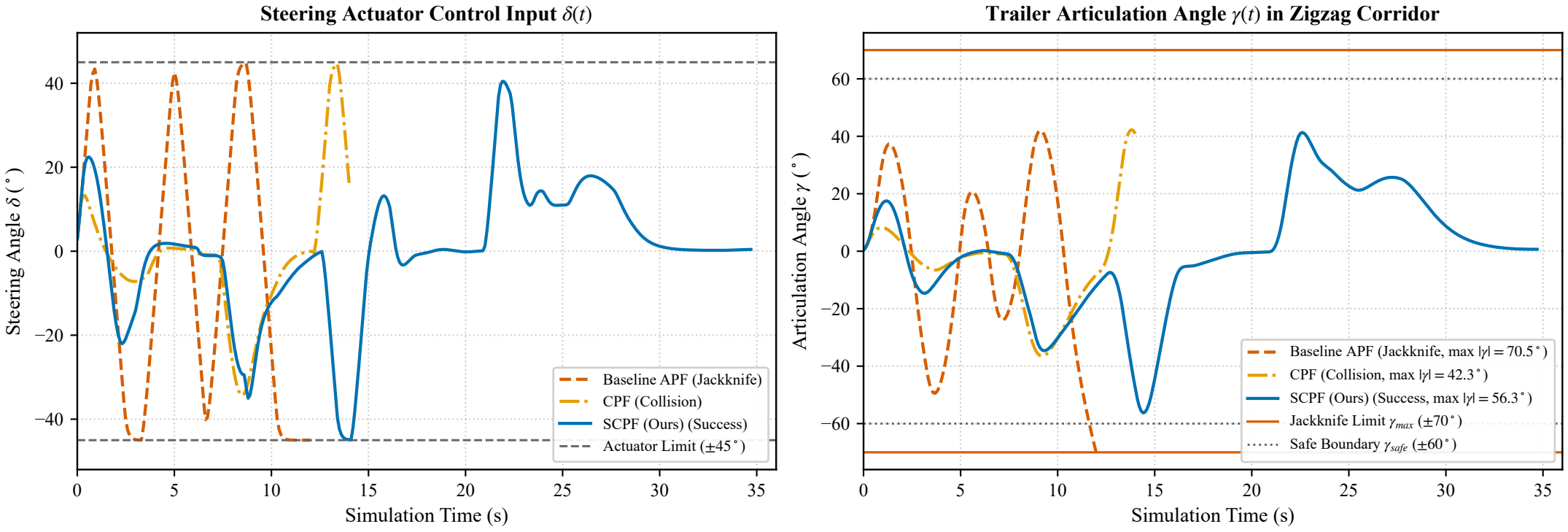

Steering Actuator Control Input $\delta(t)$
Steering Angle δ (°)
Simulation Time (s)
Baseline APF (Jackknife)
CPF (Collision)
SCPF (Ours) (Success)
Actuator Limit (±45°)
Trailer Articulation Angle $\gamma(t)$ in Zigzag Corridor
Articulation Angle γ (°)
Simulation Time (s)
Baseline APF (Jackknife, max $|\gamma| = 70.5°$)
CPF (Collision, max $|\gamma| = 42.3°$)
SCPF (Ours) (Success, max $|\gamma| = 56.3°$)
Jackknife Limit γ_{max} (±70°)
Safe Boundary γ_{safe} (±60°)